· K-d trees doesn’t work well with paged memory, they are not memory locally-efficient, as points are stored in tree nodes, so close-by points won’t lie close-by memory areas.

10.1.1 A New (More Complex) Example

To illustrate a practical situation where k-d trees are not the recommended solution, let’s pivot on our warehouse search and imagine a different scenario, where we fast forward 10 years: our e-commerce company evolved, doesn’t sell just groceries anymore, but also electronics and clothes; it’s almost 2010, and customers expect valuable recommendations when they browse our catalog – but even more important, the company’s marketing department expects that you, as CTO, makes sure to increase sales by showing to customers suggestions they actually like.

For instance, if customers are browsing smartphones (the hottest product in catalog, ramping up to rule the world of electronics, back in the day!), your application is supposed to show them more smartphones in a similar price/features range; if they are looking at a cocktail dress, they should see more dresses that looks similar to the one they (possibly) like.

Now, these two problems look (and, partially, are) so different, but they both boil down to the same core issue: given a product with a list of features, find one or more products with similar features. Obviously, the way we extract